Contents

Programming Project Phase 4 Rubric — Go-Back-N over an Unreliable UDP Channel

Total: 100 points

Requirements

Phase 4(a): Go-Back-N protocol implementation

- **R1**: Implement file transfer over UDP using **Go-Back-N (GBN)**.
- **R2**: Use a fixed sender window size `N`.
- **R3**: The sender must support **pipelined transmission** and maintain a buffer of outstanding, unacknowledged packets.
- **R4**: Implement sequence numbers, a checksum for error detection, ACKs, and a countdown timer.
- **R5**: Sender behavior must follow GBN semantics, including cumulative ACK handling.
- **R6**: On timeout, the sender must retransmit the oldest unacknowledged packet and all later packets currently in the active window (Go-Back-N recovery behavior).
- **R7**: The GBN receiver must enforce in-order delivery behavior consistent with the protocol.

Phase 4(b): Required transfer scenarios

You must demonstrate all five scenarios below:

- **R8**: Option 1 — No loss and no bit-errors.
- **R9**: Option 2 — ACK packet bit-error injection at the sender-side receive path (and correct recovery).
- **R10**: Option 3 — DATA packet bit-error injection at the receiver-side receive path (and correct recovery).
- **R11**: Option 4 — ACK packet loss injection at the sender-side receive path (and correct recovery).
- **R12**: Option 5 — DATA packet loss injection at the receiver-side receive path (and correct recovery).
- **R13**: You must be able to demo Phase 4 with and without loss recovery enabled.

Phase 4(c): Performance evaluation and plots

- **R14**: Measure end-to-end completion time for transferring the same file under each option.
- **R15**: Produce **Chart 1** for completion time versus loss/error rate from 0% to 95% in increments of 5%.
- **R16**: Each plotted point must be an average of at least 5 runs at that rate / configuration.
- **R17**: Remove (or disable) debug/print logging during timing experiments.
- **R18**: Produce **Chart 2** for completion time at a fixed 10% loss probability while varying window size.
- **R19**: Produce **Chart 3** comparing Phase 1, Phase 2, Phase 3, and Phase 4 using the same transfer file and fixed loss / bit-error settings.
- **R20**: Observe and conclude whether there is an optimal window size and an optimal timeout value for the project.

Plot requirements (what it should look like):

- **Chart 1 x-axis**: loss/error rate (%), using this list exactly:
  - `rates = [0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95]`
- **Chart 1 y-axis**: completion time in seconds (average of runs).
- **Chart 1 final plot**:
  - five lines on the same chart (Option 1, Option 2, Option 3, Option 4, Option 5), all against the same `rates` x-axis.
- **Chart 2 x-axis**: window size, using this list exactly:
  - `window_sizes = [1, 2, 5, 10, 20, 50]`
- **Chart 2 y-axis**: completion time in seconds at fixed 10% loss probability.
- **Chart 3 x-axis**: `["Phase 1", "Phase 2", "Phase 3", "Phase 4"]`
- **Chart 3 y-axis**: completion time in seconds using the same transfer file and fixed loss / bit-error settings for all phases.

Recommended data structure for averaging:

- Store raw runs in a matrix (list of lists) per option for Chart 1:

- `times_option1[r][k]` = completion time for rate index `r` and run index `k`
  - shape: `len(rates) x 5` (or more runs)
- Then compute:
  - `avg_option1[r] = mean(times_option1[r])`
  - Same for Options 2, 3, 4, and 5
- For Chart 2:
  - `times_window[w][k]` = completion time for window-size index `w` and run index `k`
  - `avg_window[w] = mean(times_window[w])`
- For Chart 3:
  - `times_phase[p][k]` = completion time for phase index `p` and run index `k`
  - `avg_phase[p] = mean(times_phase[p])`

Deliverables and submission

- **R21**: Submit `DESIGN_DOCUMENT.pdf` or `DESIGN_DOCUMENT.md`.
  - If submitting on Blackboard, it must be PDF.
  - If submitting via GitHub only, Markdown is allowed.
- **R22**: Submit `README.pdf` or `README.md` with run instructions for all scenarios, window-size experiments, and how to reproduce plots.
- **R23**: Submit the plot charts (and include the raw timing data file(s) if you generated them).
- **R24**: Submit `contribution.txt` or `contribution.doc` (individual, confidential).
- **R25**: Demo video:
  - Submit a short YouTube (unlisted) or video link showing the Phase 4 implementation working and a quick look at the generated plot(s).
- **R26**: Academic integrity: submissions may be inspected using plagiarism detection software.

Grading breakdown

A) Design Document (required first) — 15 points

- **15**: Clear, complete design submitted before major coding. Includes:
  - Sender/receiver logic aligned with GBN semantics
  - Packet format (seq number, checksum, payload, ACK format)
  - Window management / buffering strategy
  - Timeout and retransmission design
  - Error and loss injection approach for Options 2, 3, 4, and 5
  - Basic test plan and validation steps
- **10**: Mostly complete but missing one key element
- **5**: Thin/unclear or looks written after implementation

- **0**: Not submitted

B) Go-Back-N Core Correctness — 30 points

- **30**: Correct behavior across normal operation, pipelined sending, cumulative ACK handling, and timeout-based Go-Back-N recovery; no protocol logic issues
- **24**: Mostly correct with minor issues (rare edge cases or minor logic inconsistencies)
- **16**: Partially correct (protocol breaks in at least one common case)
- **0–8**: Not working / cannot be demonstrated

C) Required Scenarios (Options 1–5) — 20 points

- **20**: All five options implemented and demonstrated correctly
- **16**: Four options correct
- **12**: Three options correct
- **8**: Two options correct
- **4**: One option correct
- **0**: No correct scenario handling

D) Performance Evaluation, Plots, and Analysis — 15 points

- **15**: All required charts are present, correctly labeled, use the required x/y axes, average repeated runs properly, and include a reasonable discussion of window size / timeout behavior
- **10**: Minor issues (missing a few points, unclear labels, incomplete averaging evidence, or shallow analysis)
- **5**: Incomplete or incorrect methodology
- **0**: Missing

E) README & Reproducibility — 10 points

- **10**: Clear instructions to run all options, reproduce the charts, vary window size, and interpret outputs
- **6**: Instructions present but missing steps or ambiguous
- **3**: Minimal
- **0**: Missing

F) Contributions and Demo Video — 10 points

- **10**: Contribution file submitted and demo video clearly shows the Phase 4 implementation and plot results
- **6**: Contribution file is present but weak and/or video is incomplete
- **3**: Minimal evidence of collaboration and/or unclear demo
- **0**: Missing contribution file and/or missing video

Grader notes template

Item	Max Pts	Earned	Notes / Deductions
A (Design Document)	15		
B (Go-Back-N Core Correctness)	30		
C (Options 1–5)	20		
D (Performance, Plots, Analysis)	15		
E (README & Reproducibility)	10		
F (Contributions and Demo Video)	10		
TOTAL	**100**		